S2D-99: Best Practice Summary

> **Series:** S2D — Splunk to Dynatrace Migration | **Notebook:** 99 | **Created:** March 2026 | **Last Updated:** 10/14/2026

Overview

This notebook consolidates every actionable best practice from the S2D (Splunk to Dynatrace) migration series into a single definitive reference. Each practice specifies the exact setting, value, or action to apply.

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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS with Grail enabled
Permissions	`logs.read`, `settings.write`, `documents.write`, `automation.write`
Splunk Access	Ability to view existing queries, alerts, dashboards
Knowledge	Familiarity with both Splunk SPL and Dynatrace DQL

1. Migration Planning

#	Best Practice	Recommended Setting/Value	Priority	Category
1	Follow the 5-phase migration sequence	Phase 1: Discovery, Phase 2: Data Validation, Phase 3: Query Translation, Phase 4: Alert Migration, Phase 5: Dashboard Migration	Critical	Process
2	Inventory all Splunk assets before starting	Document every dashboard, alert, report, and saved search with their SPL queries	Critical	Process
3	Prioritize by criticality	Migrate production monitoring first, then staging, then dev	Recommended	Process
4	Map Splunk indexes to Dynatrace buckets	Each Splunk index maps to a Grail bucket; document the mapping before translation	Critical	Process
5	Validate log availability before query translation	Run DQL validation queries (S2D-02) before attempting SPL-to-DQL conversion	Critical	Process
6	Compare log volumes between platforms	Run equivalent time-bounded queries in both Splunk and Dynatrace to confirm counts match	Recommended	Validation

2. Log Discovery and Validation

#	Best Practice	Recommended Setting/Value	Priority	Category
7	Set log ingest rules at the host-group level	Use host-group scope, not per-host	Recommended	Configuration
8	Enable OneAgent log content access	`oneagentctl --set-app-log-content-access=true` on every monitored host	Critical	Configuration
9	Configure custom log sources for non-standard paths	Settings > Log Monitoring > Custom log sources; specify the exact file path pattern	Recommended	Configuration
10	Configure timestamp and splitting rules for multi-line logs	Settings > Log Monitoring > Timestamp and splitting; set the timestamp pattern and line separator	Recommended	Configuration
11	Validate by host name first, then by log source path	Use `matchesPhrase(host.name, "...")` then `matchesPhrase(log.source, "...")`	Recommended	Validation

#	Best Practice	Recommended Setting/Value	Priority	Category
12	For Kubernetes, validate by cluster then namespace then deployment	Filter chain: `k8s.cluster.name` -> `k8s.namespace.name` -> `k8s.deployment.name`	Recommended	Validation
13	Remember: host-level OneAgent settings override environment settings	If log monitoring is disabled on the OneAgent, environment-level ingest rules will not apply	Critical	Configuration

3. Query Translation (SPL to DQL)

#	Best Practice	Recommended Setting/Value	Priority	Category
14	Use `==` for equality, not `=`	`filter field == "value"`	Critical	Syntax
15	Use double quotes only, not single quotes	`"value"` not `'value'`	Critical	Syntax
16	Use curly-brace arrays, not parentheses	`in(field, {"a", "b"})` not `field IN ('a', 'b')`	Critical	Syntax
17	Translate `search index=...` to `fetch logs`	`fetch logs, from:-1h` with explicit time range	Critical	Syntax
18	Translate `where` to `filter`	`filter field == "value"`	Critical	Syntax
19	Translate `stats` to `summarize`	`summarize count = count(), by: {field}`	Critical	Syntax
20	Translate `eval` to `fieldsAdd`	`fieldsAdd new_field = expression`	Critical	Syntax
21	Translate `table` to `fields`	`fields field1, field2`	Recommended	Syntax
22	Translate `sort -field` to `sort field desc`	Explicit `asc` / `desc` keyword	Recommended	Syntax
23	Translate `head N` to `limit N`	`limit 100`	Recommended	Syntax
24	Translate `rename old AS new` to `fieldsRename`	`fieldsRename old, alias:new`	Recommended	Syntax

#	Best Practice	Recommended Setting/Value	Priority	Category
25	Translate `timechart span=5m count by level` to `makeTimeseries`	`makeTimeseries count = count(), by: {loglevel}, interval:5m`	Critical	Syntax
26	Translate `rex` field extraction to `parse` with DPL	`parse content, "DATA? 'user=' WORD:username"` (lead with `DATA?`: `parse` anchors at the field start)	Recommended	Syntax
27	Translate `*value*` wildcard to `contains()`	`contains(field, "value", caseSensitive:false)`; `matchesPhrase()` matches whole words only	Critical	Syntax
28	Translate `field IN ("a","b")` to `in()` function	`in(field, {"a", "b"})`	Critical	Syntax
29	Translate `NOT field="value"` to `filterOut` and `field!="value"` to `filter`	`filterOut matchesValue(field, "value")` keeps records without the field, like Splunk `NOT`; `filter not(matchesValue(field, "value"))` drops them, like Splunk `!=`. `matchesValue` is case-insensitive, like Splunk search	Recommended	Syntax
30	Always alias aggregation results	`summarize count = count()` not `summarize count()`	Critical	Syntax

4. Alert Migration - Anomaly Detectors

#	Best Practice	Recommended Setting/Value	Priority	Category
31	Apply the threshold translation formula	`Splunk Threshold = DT Threshold x DT Violating Samples`	Critical	Alerting
32	Use the "Alert Reimagined" strategy as default	Threshold: calculated, Sliding Window: match Splunk timeframe, Violating Samples: 2-3, Dealerting Samples: how many back-to-normal samples close the event	Recommended	Alerting
33	Set Anomaly Detector query interval to 1 minute	`interval:1m` in `makeTimeseries`	Critical	Alerting
34	Set the sliding window to match Splunk query timeframe	Max: 60 minutes. If Splunk timeframe is 15m, set sliding window to 15	Critical	Alerting

#	Best Practice	Recommended Setting/Value	Priority	Category
35	Do not map Splunk suppression onto dealerting samples	Dealerting controls when an open event closes; Splunk suppress controls re-notification. Handle re-notification in the routing workflow	Recommended	Alerting
36	Use `by: {dt.entity.cloud_application}` for entity association on logs	Ties alerts to the workload entity. The field is `deprecated` in the semantic dictionary, but its Smartscape successors (`dt.smartscape.k8s_deployment` and siblings) are not on log records yet (10/14/2026); switch once they are	Recommended	Alerting
37	Prefer Anomaly Detectors over Workflows when timeframe is 60 min or less	Per-minute evaluation and Davis correlation; queries are billed, so use Delay to reduce executions	Critical	Alerting
38	If alert timeframe exceeds 60 minutes, use Workflows or ArrayMovingSum	Anomaly Detectors max sliding window = 60 minutes	Critical	Alerting
39	Use the data-driven strategy when historical data is available	Analyze per-minute samples (24 h or less per query at 1-minute resolution; repeat for several days)	Recommended	Alerting
40	Validate the alert query returns timeseries data before configuring	Run the query in a notebook first; confirm `makeTimeseries` output	Critical	Validation

5. Alert Migration - Workflows

#	Best Practice	Recommended Setting/Value	Priority	Category
41	Use Workflows only when: timeframe > 60 min, alert runs few times/day, business-hours only, or is actually a report	Otherwise use Anomaly Detectors	Critical	Alerting
42	Specify timeframe explicitly in the DQL query, not from dashboard	`fetch logs, from:now()-7d` in the workflow query itself	Critical	Alerting
43	Workflow queries do NOT require `makeTimeseries` output	Use `summarize` to return a single aggregated value	Recommended	Alerting

#	Best Practice	Recommended Setting/Value	Priority	Category
44	Set event timeout in the JavaScript event creation step	`timeout: 15` (minutes) to auto-expire events	Recommended	Alerting
45	Workflow events do NOT auto-close or auto-update	Design event lifecycle accordingly; events expire by timeout only	Critical	Alerting
46	Workflow events are NOT correlated by Dynatrace Intelligence	No root-cause analysis or problem grouping	Recommended	Alerting
47	Account for workflow license consumption	Every standard workflow is billed per hour it exists, regardless of how often it runs; the DQL it executes is billed separately as query consumption	Recommended	Licensing
48	For business-hours-only alerting, filter by hour in the Dynatrace Intelligence query instead of using a Workflow	`fieldsAdd hour = getHour(timestamp, timezone:"America/New_York")` then `filter hour >= 8 and hour < 18` (queries run in UTC; set your time zone)	Recommended	Alerting

6. Extended Timeframes (ArrayMovingSum)

#	Best Practice	Recommended Setting/Value	Priority	Category
49	Use `arrayMovingSum(array, 60)` with `interval:1m` for 60-minute rolling sum	Each data point = sum of preceding 60 minutes	Critical	DQL
50	Max `arrayMovingSum` window = 60	Cannot exceed 60 data points regardless of interval	Critical	DQL
51	With Anomaly Detectors + ArrayMovingSum, set sliding window = 60 and violating samples = 1	The query only sees its sliding window: with a 1-minute window each "60-minute" sample is a 1-minute count. The first 59 points of any window are partial sums. Derived, not tested on a live custom alert	Recommended	Alerting
52	Set threshold to total count expected in the rolling window	If Splunk threshold = 500 errors in 60 min, set DT threshold = 500	Critical	Alerting
53	Remove the original timeseries field after adding the rolling sum	`fieldsRemove error_count` after `fieldsAdd error_count_1h = arrayMovingSum(error_count, 60)`	Recommended	DQL

#	Best Practice	Recommended Setting/Value	Priority	Category
54	For dashboard visualizations > 60 min, increase the interval	`interval:5m` with `window:48` = 4-hour rolling sum (intervals snap to supported values; check the returned `interval`)	Recommended	DQL
55	For alerts > 60 minutes that cannot use proportional reduction, use Workflows	If error distribution is uneven, proportional threshold reduction will produce false positives	Recommended	Alerting

7. Metric Creation from Logs

#	Best Practice	Recommended Setting/Value	Priority	Category
56	Create log-based metrics for frequently executed dashboard and alert queries	Reduces query cost, improves speed, increases retention	Recommended	Performance
57	Metric filter must use only supported functions	`matchesPhrase`, `matchesValue`, `isNull`, `isNotNull`, basic operators only. No `parse` or `fieldsAdd`	Critical	Configuration
58	Avoid high-cardinality dimensions	Never use transaction IDs, session IDs, request IDs, or timestamps as metric dimensions	Critical	Configuration
59	Consolidate similar metrics using dimensions instead of separate metrics	One `service_errors` metric with `k8s.deployment.name` dimension, not per-service metrics	Recommended	Design
60	Name metrics as `log.[app_name].[metric_description]`	Example: `log.easytravel.error_count`	Recommended	Naming
61	Complete all SPL-to-DQL translation before requesting metric extraction	Metric filters must match the final DQL query filters	Critical	Process
62	Query extracted metrics with `timeseries` not `fetch logs`	`timeseries sum(log.easytravel.error_count), from:-1h` (counter: use `sum()`, not `avg()`)	Critical	Syntax

8. Dashboard Migration

#	Best Practice	Recommended Setting/Value	Priority	Category
63	Layout: top row = single-value KPIs, middle = time-series trends, bottom = detail tables	Standard 3-tier dashboard layout	Recommended	Design
64	Use dashboard variables for interactive filtering	Define `$cluster`, `$namespace`, `$deployment` variables with query-backed dropdowns	Recommended	Design
65	Populate dropdown variables with DQL	`fetch logs, from:-1h | summarize count(), by:{field} | fields field | sort field asc`	Recommended	Design
66	Use consistent time intervals across related charts on the same dashboard	All trend charts should use the same `interval:` value	Recommended	Design
67	Map Splunk gauges to the Gauge or Meter bar visualization	Radial gauge → Gauge; filler/marker gauge → Meter bar or Gauge	Recommended	Design
68	Create Log Searcher dashboards for investigation	VM version: variables for `host_name` and `log_source`. K8s version: variables for `cluster`, `namespace`, `deployment`	Recommended	Design
69	Validate migrated dashboard data against Splunk	Compare identical time ranges side-by-side before decommissioning Splunk dashboards	Critical	Validation

9. Naming Standards

#	Best Practice	Recommended Setting/Value	Priority	Category
70	Dashboard names	`[app_name] Dashboard Title`	Critical	Naming
71	Alert configuration names	`[app_name] alert name from splunk`	Critical	Naming
72	Alert event names (triggered title)	`[app_name] [priority] - {dimension} - alert name` e.g. `[EasyTravel] P2 - {dims:host.name} - High Error Count`	Critical	Naming
73	Report names	`[Report] [app_name] Report Title`	Recommended	Naming

#	Best Practice	Recommended Setting/Value	Priority	Category
74	Lookup table paths	`/lookups/[app_name]/[table_name]`	Recommended	Naming
75	Log-based metric names	`log.[app_name].[metric_description]` (lowercase, underscores)	Recommended	Naming
76	Alerting workflow names	`[Alert] [app_name] Alert Description`	Recommended	Naming
77	Automation workflow names	`[Auto] [app_name] Automation Description`	Recommended	Naming
78	Remediation workflow names	`[Remediate] [app_name] Remediation Description`	Optional	Naming
79	Enforce naming during migration, not after	Review every asset name at creation time	Critical	Process
80	Use infrastructure-as-code to enforce naming	Monaco or Terraform templates with naming prefixes built in	Recommended	Process

10. DQL Performance and Syntax

#	Best Practice	Recommended Setting/Value	Priority	Category
81	Always specify a time range on `fetch`	`fetch logs, from:-1h` (never bare `fetch logs`)	Critical	Performance
82	Filter immediately after `fetch`	Place `filter` commands before any `fieldsAdd`, `parse`, or `summarize`	Critical	Performance
83	Use `==` for exact matches, `matchesPhrase` for token search	`==` is faster than `~` or `matchesPhrase` when the full value is known	Recommended	Performance
84	Target specific Grail buckets	`fetch logs, bucket: {"app_logs_*"}` to avoid scanning all buckets	Recommended	Performance
85	Use named parameters for DQL functions	`round(value, decimals: 2)` not `round(value, 2)`	Critical	Syntax
86	Use `countIf()` instead of nested filter + count	`summarize errors = countIf(status == "ERROR")` in a single pass	Recommended	Performance

#	Best Practice	Recommended Setting/Value	Priority	Category
87	Check `isNotNull()` before aggregating optional fields	`filter isNotNull(db.system)` before `summarize avg(duration), by:{db.system}`	Recommended	Correctness
88	Sort and limit must come last in the pipeline	`sort field desc | limit 10` as final commands	Critical	Performance
89	Use `fieldsKeep` or `fieldsRemove` early to drop unneeded columns	Reduces data volume through the pipeline	Recommended	Performance
90	Use `samplingRatio:` for exploratory queries on large datasets	`fetch logs, samplingRatio:100` then multiply results back	Optional	Performance

Summary

This notebook contains **90 best practices** across 10 categories for Splunk to Dynatrace migration:

Category	Count	Priority Breakdown
Migration Planning	6	4 Critical, 2 Recommended
Log Discovery and Validation	7	2 Critical, 5 Recommended
Query Translation (SPL to DQL)	17	11 Critical, 6 Recommended
Alert Migration - Anomaly Detectors	10	6 Critical, 4 Recommended
Alert Migration - Workflows	8	3 Critical, 5 Recommended
Extended Timeframes (ArrayMovingSum)	7	3 Critical, 4 Recommended
Metric Creation from Logs	7	4 Critical, 3 Recommended
Dashboard Migration	7	1 Critical, 6 Recommended
Naming Standards	11	4 Critical, 6 Recommended, 1 Optional
DQL Performance and Syntax	10	4 Critical, 5 Recommended, 1 Optional
Total	**90**	**42 Critical, 46 Recommended, 2 Optional**

References

- **S2D-01** - Getting Started: Migration overview and planning
- **S2D-02** - Locating Logs: Data validation and ingest configuration
- **S2D-03** - SPL to DQL: Query translation fundamentals

- **S2D-04** - Anomaly Detectors: Continuous alert migration
- **S2D-05** - Workflow Alerts: Scheduled and extended alerting
- **S2D-06** - ArrayMovingSum: Extended timeframe handling
- **S2D-07** - Metric Creation: OpenPipeline log-based metrics
- **S2D-08** - Dashboard Migration: Visualization conversion
- **S2D-09** - Naming Standards: Asset organization conventions

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